

Ameya Panchal

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Education

The Pennsylvania State University

B.S. Computer Science, Minors in Physics and AI • Schreyer Honors College • GPA 3.71 / 4.00

University Park, PA

Expected May 2028

Research Experience

Do Residual-Stream Probes Survive Degrading Chain-of-Thought Legibility?

SPAR • mentored by Marios Tsatsos, with two fellow mentees • PyTorch

Sep 2026 – Present

- Finding borderline prompts to train an activation probe on, and replicating chain-of-thought illegibility
- Testing what current chain-of-thought monitors miss, before settling on the project's direction

The J-lens offset is the model's token frequency: z-scoring helps [post]

Independent • PyTorch, jacobian-lens • Qwen3, Pythia, GPT-2, Gemma-2

Aug 2026 – Sep 2026

- Traced the per-token bias in a lens that reads a large language model's internal activations as words to the model's own word frequency (Spearman 0.48); subtracting it made readouts $3.5-7\times$ worse
- Replaced subtraction with z-scoring: recovered hidden secret words from fine-tuned models where the published method scored zero, and beat it 0.805 to 0.665 on Gemma-2-9B

A Dead-ReLU Absorbing State — Resolving an Open Memorization Anomaly [LessWrong]

Independent • PyTorch • LessWrong

Jul 2026

- Explained why one small transformer in a published memorization challenge could barely learn: its ReLU units died in training and cut off the gradient, a training failure, not a capacity limit
- Confirmed it with one controlled experiment (a bias after the hidden layer restored full capacity, 1024 facts; before it, none, 488) and hand-built weights storing 672 facts; *overturned* the authors' explanation

An Induction Head in Disguise — Circuit Analysis of a Character-Level Transformer [LessWrong]

Independent • PyTorch, TransformerLens • LessWrong

Jul 2026

- Trained a 6-layer character-level transformer on Nietzsche's prose and located the one attention head that copies earlier text forward
- *Ruled out* two hypotheses about the head (quotation and parenthesis matching) with targeted controls
- Found the head had learned half of a bracket-matching mechanism: it knows what to write, but never learned where to look

Systematic Bias from Assuming Isotropic Surface Emission in Neutron-Star Pulse-Profile Modeling [poster]

Penn State Multi-Campus REU • advised by Prof. Asif ud-Doula • Python

Oct 2025 – Aug 2026

- Simulated X-ray photons scattering through a neutron star's atmosphere to measure the error a common assumption introduces
- Validated it against an exact analytic solution (reduced $\chi^2 = 0.70$) and published waveforms (0.11% peak error)
- Showed the error shifts between two observables; first-author paper under review at *Astrophysics and Space Science*

Experience

Nittany AI Advance Internship Program

Application Specialist • Team of 5 • Client: Penn State Office of the Physical Plant

State College, PA

Dec 2025 – Apr 2026

- Built a Tesseract OCR + LLM + regex pipeline that extracts metadata (project name, ID, and more) from scanned project documents
- Raised extraction accuracy up to 10% by merging the LLM and regex paths and moving from GPT-4o to GPT-5 after benchmarking Azure's 4- and 5-series models; parallelized the pipeline for 5x faster runs

Projects

CrashAI — Crash-Risk Analysis Platform

Founder & Technical Lead • Team of 3 • 1st of 40 teams, Nittany AI Challenge 2026 • Scoped with PennDOT

Feb 2026 – Present

- Built a crash-severity model (XGBoost, 81 road and driver features) with per-site explanations and what-if simulation; led a team building a 33-tool agent, retrieval over engineering manuals, and an eval harness

Skills

Machine Learning: Python, PyTorch, HuggingFace Transformers, TransformerLens, NumPy, scikit-learn, Monte Carlo simulation, Git

Mechanistic Interpretability: Circuits, OV/QK analysis, activation and path patching, direct logit attribution, Jacobian and logit lenses